

Research papers

Hillslope-scale variability of soil water potential over humid alpine forests: Unexpected high contribution of time-invariant component

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ABSTRACT

Soil water potential (SWP) gradients control water and energy fluxes in the Soil – Plant – Atmosphere Continuum. Exploring the spatial – temporal variability of SWP at the hillslope scale is an essential step towards developing ecohydrological models. In this study, SWP monitoring was carried out on two contrasting, alpine, humid forested hillslopes (i.e., coniferous and broadleaf forests) at three depths (10, 50, and 100 cm). The effects of static (soil properties and topography) and dynamic (precipitation dynamics and soil wetness conditions) factors on SWP dynamics and variability were investigated based on statistical methods, temporal stability, and variance decomposition analysis. The results showed that the SWP in the soil – root zone (10 and 50 cm depths) in the broadleaf forest exhibited higher variability and lower temporal stability than in the coniferous forest, likely because of the stronger evapotranspiration and more variable soil hydraulic conductivity. The relatively high SWP variability in the deeper soil layer (100 cm depth) of both forested hillslopes was due to the frequent lateral/preferential flow. Furthermore, topographic factors (i.e., distance of hillslope to stream and slope) and soil properties (clay and rock fragment contents) regulated the spatial patterns in SWP and influenced representative site selection in the coniferous and broadleaf forests, respectively. Decomposition of the spatial variance showed that the time-invariant component was the dominant contributor to the total spatial variance. An exponentially increasing relationship between the time-invariant component and mean SWP and a concave-up parabolic shape between the contribution of the temporal anomalies and mean SWP were observed. Our results provide an improved understanding of the temporal and spatial variability in terms of SWP in forested hillslopes.

1. Introduction

Soil moisture is a key variable that controls the exchange of water and energy between the soil, vegetation, and climate (Guo and Lin, 2016; Penna et al., 2009; Seneviratne et al., 2010; Vereecken et al., 2022). However, soil moisture is often spatially and temporally heterogeneous due to the complex interaction of precipitation, vegetation, soil properties, and topography (Teuling and Troch, 2005; Vereecken et al., 2014; Wang et al., 2017; Western et al., 2002). Knowledge of this spatial–temporal variability is essential for various scientific investigations, including (1) uncertainty estimations of the hydrological

models (Heathman et al., 2003), (2) optimal location of sites for soil moisture monitoring and sampling (Wang et al., 2008), (3) validation of remote sensing products (Vereecken et al., 2014; Vereecken et al., 2007).

Previous studies have shown that the spatial–temporal variability of soil moisture can be characterized in terms of the standard deviation or coefficient of variance of soil moisture as a function of its mean value (Famiglietti et al., 2008). However, there is no consensus regarding the documentation of soil moisture variability at different spatial scales and in variable climate zones (Korres et al., 2015). Besides, because of practical limitations, conclusions regarding soil moisture variability are

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based almost solely on the near-surface of gentle slopes (e.g., see the reviews in Brocca et al., 2007; Vanderlinden et al., 2012). Steep hillslopes and deep soil layers in alpine humid forests, have complex subsurface critical zone structures and frequent rainfall pulses that may severely affect the soil moisture distribution (Guo and Lin, 2016; Lin, 2006; Lin and Zhou, 2008; Wang et al., 2022; White et al., 2019). Grayson and Western (2001) suggested that the spatial patterns of soil moisture were more related to topographic variables in humid and steep hillslopes, where the subsurface lateral flow is likely to occur. Lin (2006) introduced a conceptual model that considered the significant effects of the preferential/lateral flow system on the spatial soil moisture distribution within soil profiles. Kaiser and McGlynn (2018) identified an increased standard deviation of the water content within a riparian area caused by the effects of lateral flow from connected flow paths compared with that in the uplands area. However, the variability of soil moisture in steep and humid hillslope scale is still inadequate and requires further investigation.

The temporal stability analysis method proposed by Vachaud et al. (1985) has been used as a diagnostic tool to identify time-stable locations and potential controlling factors. The temporal stability of soil moisture at specific locations was shown to be consistently wetter or drier than the average value, leading to a persistent bias, which is a reflection of the temporal persistence of the spatial structure (Martínez-Fernández and Ceballos, 2005). This method has been successfully tested in different vegetation types, soil properties, topographic attributes, soil depths, and seasonal soil wetness conditions (Baroni et al., 2013; Gao et al., 2011; Hu et al., 2010; Lin, 2006; Martínez-Fernández and Ceballos, 2005; Wang et al., 2015; Williams et al., 2009; Zhu et al., 2021). Cho and Choi (2014) found that the meteorological factors contributed more to soil moisture temporal stability on a regional scale. Zhu et al. (2021) showed that throughfall, soil bulk density, and litter biomass significantly affected the soil moisture temporal stability at a plot scale. Moreover, temporally stable locations identified give the areal mean soil moisture of a hillslope rather than sampling at many random points (Jacobs et al., 2004), which highlights the usefulness of temporal stability analysis.

The controls on spatial and temporal soil moisture patterns are a combination of static (e.g., topography and soil properties) and dynamic (e.g., precipitation and soil wetness conditions) variables. As such, an analysis of the variability of soil moisture patterns can be carried out by considering the absolute soil moisture values (expressed in volumetric terms; cm³/cm³) or relative values, such as temporal anomalies, which reduces the impact of the static soil properties (Brocca et al., 2014; Mittelbach and Seneviratne, 2012; Yetbarek and Ojha, 2020). Similarly, the variability of soil moisture can be decomposed into time-invariant (i.e., temporal mean) and time-variant (i.e., anomalies) components. It is possible to assess the extent to which these components contribute to the overall spatial soil moisture variability. Brocca et al. (2014) showed that there was much lower spatial variability of temporal anomalies than absolute soil moisture values based on six soil moisture monitoring networks across six countries, indicating a major contribution of the time-invariant component. Gao et al. (2015) found that the time-invariant component provided the primary contribution and varied with soil wetness. As such, it is important to examine how these components vary as a function of soil wetness conditions, as soil moisture variability changes significantly with soil wetness. Moreover, previous studies have focused mainly on regional or catchment scales and near-surface soil moisture dynamics, and studies of deeper soil layers at different spatial scales are encouraged and required (Dari et al., 2019; Gao et al., 2015; Mittelbach and Seneviratne, 2012; Rötzer et al., 2015).

Soil water potential (SWP) drives water fluxes between soils, streams, and groundwater, but the relationship between SWP and variability is controversial due to the nonlinear relationship between SWP and soil moisture. In addition, the studies of SWP variability are rather limited compared with the soil moisture (Novick et al., 2022; Yang et al., 2016; Yu et al., 2015). Therefore, this study investigated the

SWP behaviors at the hillslope scale, in which topography and rainfall inputs are expected to have an important role. The specific objectives were (1) to investigate spatial-temporal variability of SWP over contrasting forested hillslopes; (2) to evaluate the contribution of the time-invariant and -variant component in the determination of the spatial variability of SWP; and (3) to explore how soil wetness conditions alter contribution of the time-invariant and -variant components. The results of this study can be of interest mainly for small-scale studies (e.g., hydrological or ecological studies) and provide guidelines for the determination of sampling or monitoring schemes.

2. Materials and methods

2.1. Sites description and data collection

The Hailuo Valley is located on the eastern slope of Mount Gongga, on the eastern edge of the Tibetan Plateau. The elevation increases dramatically over a short horizontal distance, resulting in contrasting land cover along the elevation gradient. The climate is characterized by distinct dry (November–April) and wet (May–October) seasons, and almost 80 % of the total rainfall occurs during wet seasons (Hu et al., 2018; Sun et al., 2011). This study focuses on a coniferous forest (3020 m asl) dominated by *Abies fabri*, *Picea brachytyla*, and sparse shrubs and a broadleaf forest (2280 m asl) dominated by *Lithocarpus cleistocarpus* and *Cercidiphyllum japonicum*. The mean annual precipitation and temperature in the study area are 1884.2 mm and 3.98 °C in the coniferous forest and 1572.4 mm and 8.59 °C in the broadleaf forest, based on the nearby rainfall and temperature monitoring devices (i.e., within a horizontal distance of 500 m), which were attached to an eddy covariance flux tower (Hu et al., 2018).

For each forest type, a hillslope near a stream was chosen to deploy the SWP sensors (T4e; UMS Corp, Germany; −85 kPa to 100 kPa) at three depths (10, 50, and 100 cm) and three slope positions (upper slope [U], middle slope [M], and lower slope [L]), with two replicates (we used 1 and 2 to denote the replicate in the left and right column, for example, U1 denotes the monitoring location at left column on the upper hillslope), and the horizontal distance remained constant between the replicates along the hillslope. The relief of the two hillslopes are convergent, with average slopes of 20° and 26° and areas of 0.40 and 0.36 ha in the coniferous and broadleaf forests, respectively. The hillslopes are devoid of human activity and the soils are primarily sandy (sand > 90 %) with abundant rock fragments (Table S1). Along the both hillslopes, the lower boundary of dense plant roots is mainly at 40–60 cm, above which the soils have a high infiltration capacity and highly variable root networks. Below the root boundary, rock fragments are more common, and the infiltration capacity decreases abruptly (Wang et al., 2022).

Five throughfall collection trenches were used to synchronously measure the throughfall (i.e., rainfall after canopy interception) and were evenly distributed on the hillslopes to reduce the spatial variability of the throughfall. All hydrological monitoring equipment was connected to a data logger (CR1000; Campbell Scientific Inc, USA). The data were recorded every 15 min and unreliable data were excluded during relatively dry seasons (November–April with frozen soils), battery damage, and maintenance periods.

2.2. Statistical and temporal stability analysis

Statistical features of the spatial and temporal variability of the SWP were determined. The standard deviation σ_j and coefficient of variability in space CV_j were the basis of the calculations. We defined φ_{ij} , which denotes the SWP measured at location i ($i = 1, 2, 3, \dots, n$) on the monitoring day j ($j = 1, 2, 3, \dots, m$). The spatial mean of each monitoring day $\bar{\varphi}_j$ was calculated as follows (to simplify the notations, the depth k was omitted in the following expressions):

$$\bar{\varphi}_j = \frac{1}{n} \sum_{i=1}^n \varphi_{ij} \quad (1)$$

In a similar way, the temporal mean value for each monitoring location σ_j was defined as follows:

$$\bar{\varphi}_i = \frac{1}{m} \sum_{j=1}^m \varphi_{ij} \quad (2)$$

The coefficient of variance in space CV_j was obtained as follows:

$$CV_j = \frac{\sigma_j}{\bar{\varphi}_j} = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (\varphi_{ij} - \bar{\varphi}_j)^2}}{\bar{\varphi}_j} \quad (3)$$

where σ_j is the spatial standard deviation, which reflects the spatial variability of the SWP.

The temporal stability analysis was carried out using the relative difference method. The relative difference (RD) between the φ_{ij} at location i on time j and the spatial mean $\bar{\varphi}_j$ at the same time was defined as follows:

$$RD_{ij} = \frac{\varphi_{ij} - \bar{\varphi}_j}{\bar{\varphi}_j} \quad (4)$$

The mean relative difference (MRD) for location i was then calculated by:

$$MRD_i = \frac{1}{m} \sum_{j=1}^m RD_{ij} \quad (5)$$

The standard deviation of the MRD was calculated as follows:

$$SDRD_i = \sqrt{\frac{1}{m-1} \sum_{j=1}^m (RD_{ij} - MRD_i)^2} \quad (6)$$

The time-stable locations tend to have a MRD value close to zero and a minimum SDRD, as the MRD and SDRD represent the bias and precision when predicting the mean SWP. Thus an introduced index of time stability, ITS, that considers the bias and precision (Gao et al., 2020; Jacobs et al., 2004; Zhao et al., 2010) was calculated as follows:

$$ITS_i = \sqrt{MRD_i^2 + SDRD_i^2} \quad (7)$$

The ITS is a single metric for identifying sampling locations that are most representative of the mean hillslope SWP. The lowest ITS represents the most stable locations and vice versa.

To assess how the temporal stability indexes depend on different variables, a Pearson Correlation Analysis was used to quantify the dependency of the temporal stability indexes (i.e., MRD and ITS) on associated soil and terrain features. Soil properties and hillslope topography has been widely proven to control the temporal stability of the SWP in previous studies (Vanderlinden et al., 2012). Soil properties, such as bulk density (BD), clay, sand, and silt contents, soil organic matter (SOM), saturated hydraulic conductivity (K_s), and rock fragment content (RF), reflect the ability of soil to retain and transmit water, and is the reason for differences in the SWP. While, in the hillslope area with less soil variability, the influence of topographic features on SWP variation may increase, hence the slope and distance to stream (DTS) were considered for further analysis.

2.3. Decomposition of spatial variance

Based on the theory introduced by Mittelbach and Seneviratne (2012), the spatial variance of the SWP, $\sigma^2(\varphi_{ij})$, can be decomposed into three components:

$$\sigma^2(\varphi_{ij}) = \sigma^2(\bar{\varphi}_i) + \sigma^2(A_{ij}) + 2cov(\bar{\varphi}_i, A_{ij}) \quad (8)$$

where $\sigma^2(\bar{\varphi}_i)$ is the spatial variance of the temporal mean SWP, $\sigma^2(A_{ij})$ is the spatial variance of the temporal anomalies, and $2cov(\bar{\varphi}_i, A_{ij})$ is the spatial covariance between the temporal mean SWP at a specific location and its corresponding anomaly.

A_{ij} represents the temporal anomalies and was calculated as follows:

$$A_{ij} = \varphi_{ij} - \bar{\varphi}_i \quad (9)$$

In Eq (8), $\sigma^2(\bar{\varphi}_i)$ is the time-invariant component, whereas the sum of $\sigma^2(A_{ij})$ and $2cov(\bar{\varphi}_i, A_{ij})$ which varies over time, is the time-variant component. This allows us to determine the percentage contribution of the individual components to the total variance and their temporal evolution at different soil depths.

3. Results

3.1. Overview of soil water potential dynamics

The time series of the mean SWP observed on the hillslopes and its responses to throughfall input are shown in Fig. 1. The SWP responded rapidly to throughfall input and then decreased significantly at the end of the throughfall in both forests (Fig. 1). On days with no rain, the SWP at 50 cm depth exhibited a significant decrease compared with that of the surface soil layer in both forests. A summary of $\bar{\varphi}_j$ during the wet seasons for different soil depths is shown in Table 1, with mean values of -26.69 , -16.13 , and -10.63 hPa in the coniferous forest and -21.19 , -22.22 , and -12.50 hPa in the broadleaf forest at 10, 50, and 100 cm depths, respectively. The range of the $\bar{\varphi}_j$ at 10, 50, and 100 cm depths was 70.60, 32.29, and 25.39 hPa in the coniferous forest and 53.19, 53.84, and 28.93 hPa in the broadleaf forest, respectively (Table 1). During wet seasons, the more converged shapes and unimodal distribution of the SWP distribution in the coniferous forest are indicative of less variable SWP dynamics (Fig. 2). Notably, in the study period, the occurrence of $SWP \geq 0$ was observed at all monitoring locations, indicating the ephemeral occurrence of perched water tables. It also showed that the percentage duration of the soil saturation status was similar between both forests (36.25 % in the coniferous forest and 35.47 % in the broadleaf forest). The soil layer at 100 cm depth on the lower slope in the coniferous forest was saturated for nearly a third of the study period.

In this study, the standard deviation of the SWP decreased with increasing mean SWP except for the value for 100 cm depth in the broadleaf forest (Fig. 3), with mean standard deviation values of 7.28 hPa with range 1.72–24.48 hPa, 6.13 hPa with range 2.75–12.28 hPa and 8.10 hPa with range 4.03–12.58 hPa in the coniferous forest and 7.93 hPa with range 0.99–21.96 hPa, 12.00 hPa with range 1.12–32.04 hPa and 7.69 hPa with range 2.34–16.60 hPa in the broadleaf forest from surface to subsurface soil depths (Table 1). The CV_j was 0.42 with range 0.12–1.03, 0.59 with range 0.25–1.13, and 0.82 with range 0.49–3.28 in the broadleaf forest and 0.31 with range 0.07–1.56, 0.41 with range 0.08–3.00, and 0.96 with range 0.18–4.67 in the coniferous forest from surface to subsurface soil depths, indicative of more variable SWP dynamics in the soil-root zone in the broadleaf forest (Table 1).

3.2. Temporal stability analysis and controlling factors

The temporal stability analysis was carried out by calculating the MRD and ITS of the mean SWP on the hillslopes at different monitoring locations and depths (Fig. 4a–c and 5a–c). The analysis identified the most representative locations where SWP measurements should be conducted, or continuous monitoring sensors should be installed for estimating the temporal pattern of the hillslope mean SWP and explored potential factors that control the temporal stability of the SWP.

The characteristics of the rank-ordered MRD for each monitoring location at the three soil depths differed in both forests (Figs. 4–5). The

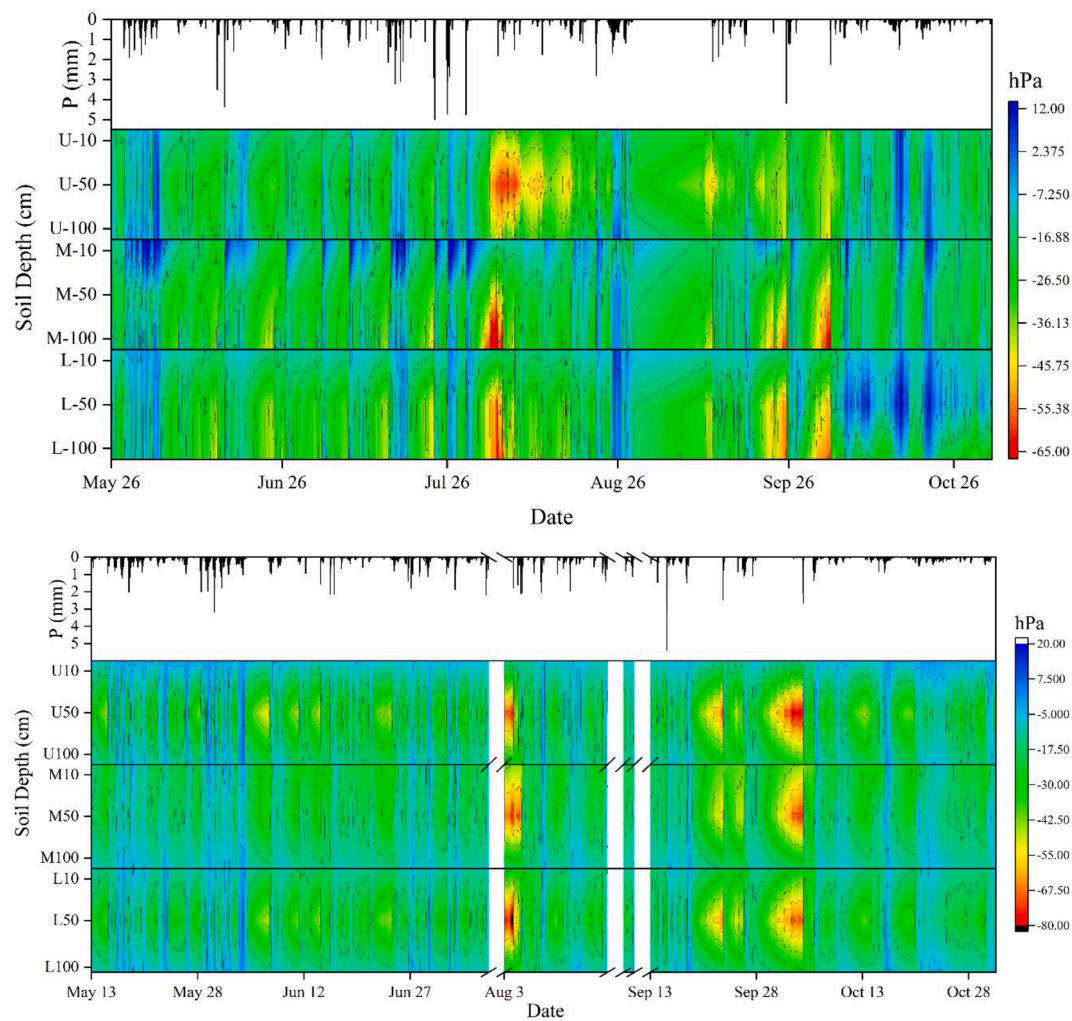


Fig. 1. Dynamics of the soil water potential (SWP) and throughfall in the broadleaf (up) and coniferous forest (down) across wet seasons 2021 at different soil depths and slope positions (U, M, and L denotes upper slope, middle slope, and lower slope, respectively).

Table 1

Summary of spatial mean soil water potential statistics over the two forested hillslope for the various soil depths.

Summary statistics	Coniferous forest			Broadleaf forest		
	10 cm	50 cm	100 cm	10 cm	50 cm	100 cm
Mean (hPa)	-26.69	-16.13	-10.63	-21.19	-22.22	-12.50
Median (hPa)	-22.76	-15.31	-10.77	-20.28	-22.70	-13.44
Standard deviation (hPa)	7.28	6.13	8.10	7.93	12.00	7.69
Coefficient of variance	0.31	0.41	0.96	0.42	0.59	0.82
Inter-quart. range	14.41	8.07	8.23	11.23	13.91	6.85
Maximum value (hPa)	-6.40	-5.20	0.92	-4.11	1.95	3.36
Minimum value (hPa)	-77.00	-37.49	-24.47	-57.30	-51.89	-25.57

ranges for the MRD at 10, 50, and 100 cm depths were 63.5 %, 99.1 %, and 193.7 % in the coniferous forest and 86.2 %, 160.1 %, and 93.8 % in the broadleaf forest, respectively. The mean value of ITS was 0.30 with range 0.20–0.47 and 0.33 with range 0.12–0.74 in the coniferous forest and 0.39 with range 0.27–0.47 and 0.65 with range 0.45–1.15 in the broadleaf forest at 10 and 50 cm depths, respectively. However, the ITS values at 100 cm depth were considerably larger, with a mean value of

2.64 with range 0.86–5.57 in the broadleaf forest and 2.67 with range 1.20–5.22 in the coniferous forest.

The most representative locations, in terms of the lowest ITS values, were M1 at 10 cm depth, L1 at 50 cm depth, and M2 at 100 cm depth in the coniferous forest. Notably, the representative location at 100 cm depth in the coniferous forest was M2, where the SWP was closest to the mean of the studied hillslope, with an R^2 value equal to 0.93 (Fig. 5f), despite the value of ITS being worse than U2. The representative locations identified were L1 at 10 cm depth, U2 at 50 cm depth, and L1 at 100 cm depth in the broadleaf forest, respectively. To test the feasibility of the identified representative locations, the measured hillslope mean SWP was plotted against the representative locations (Fig. 4d–f and 5d–f). The linear fits had high correlation coefficients ($R^2 > 0.83$) for the three depths, indicating these representative locations are suitable for estimating hillslope mean SWP at various depths.

A correlation analysis between the MRD and potential variables revealed that no factors had a controlling effect in the broadleaf forest. In contrast, the DTS and slope were identified as controlling factors in the coniferous forest (Fig. 6). In terms of the ITS, K_s was the only factor that affected the temporal stability in the coniferous forest, and soil bulk density, SOM, clay, and rock fragment content were identified as potential variables controlling the temporal stability of the SWP in the broadleaf forest.

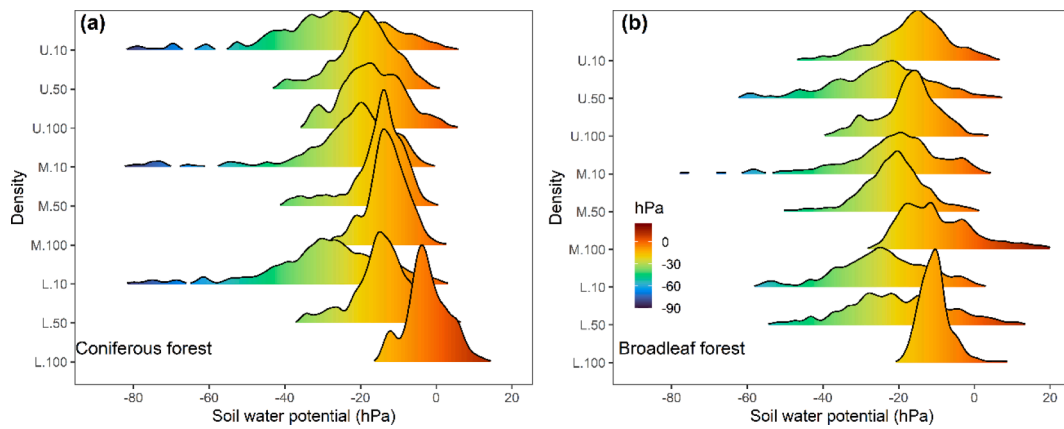


Fig. 2. Ridgeline plots of the SWP (hPa) by topographic position at different soil depths (10, 50 and 100 cm depth) in the coniferous (left) and broadleaf forests (right).

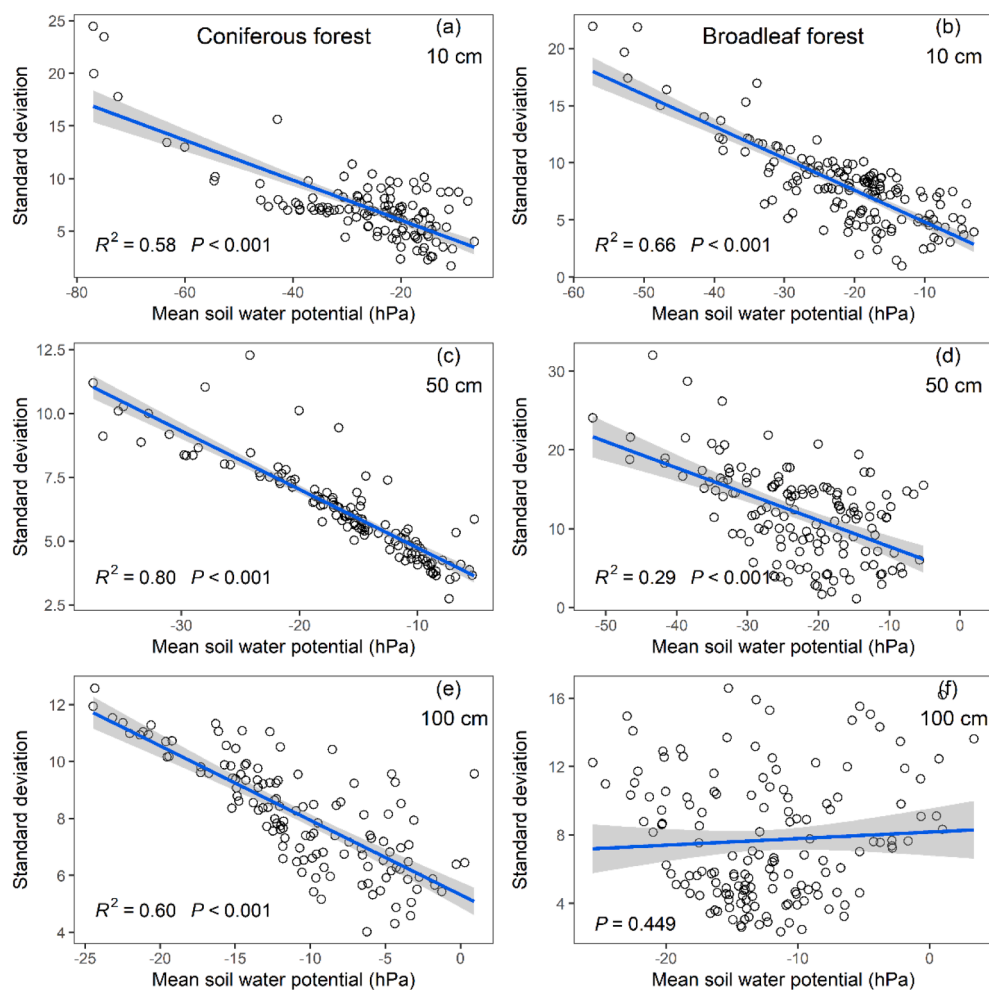


Fig. 3. Scatter plots of standard deviation of SWP against mean SWP and linear fitting (blue line, shade area denotes the fitting line and confidence level at 95%) at different depths in the coniferous (left column) and broadleaf forests (right column). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Decomposition of spatial variance

As shown in Eq (8), the spatial variance of the absolute SWP on any given date consists of the temporal mean, temporal anomalies (i.e., the absolute SWP minus the seasonal mean), and covariance. The time series of contribution of different components to the total spatial variance

during the study period are shown in Fig. 7. For the study period, the time-invariant component (i.e., temporal mean) contributed the most, with mean values of 77.84 % (2.66 %–255.07 %) in the coniferous forest and 93.22 % (8.54 %–346.88 %) in the broadleaf forest (Table 2). The contribution of the time-invariant component in the broadleaf forest was higher than that in the coniferous forest at 10 and 100 cm depths. The

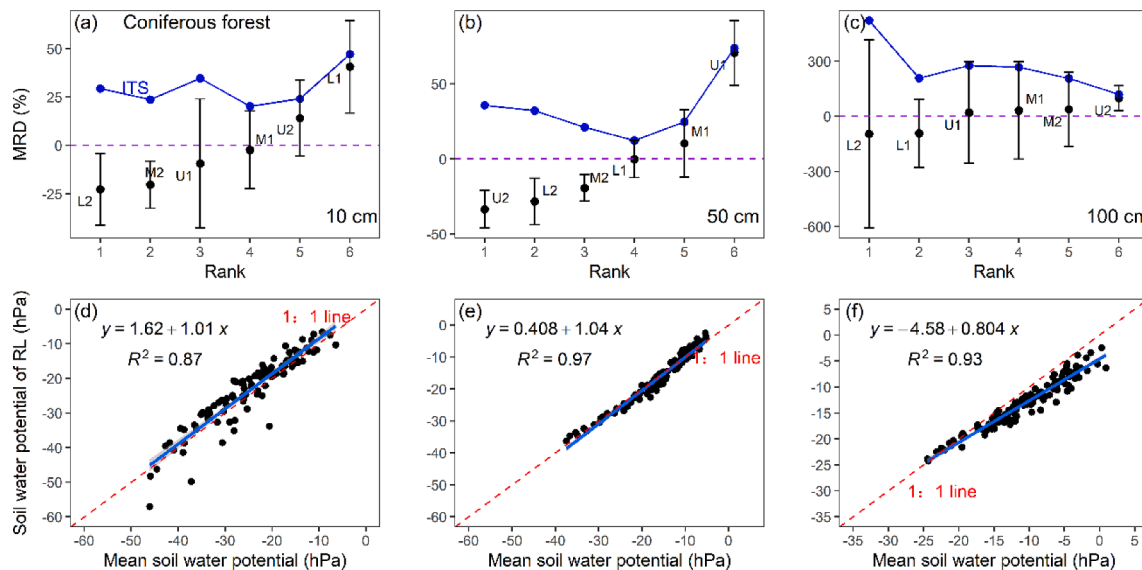


Fig. 4. Ranked MRD (with one standard deviation) for different soil depths (a, b, c), and comparison of the hillslope mean SWP versus SWP observed at the most representative locations (d, e, f) in the coniferous forest.

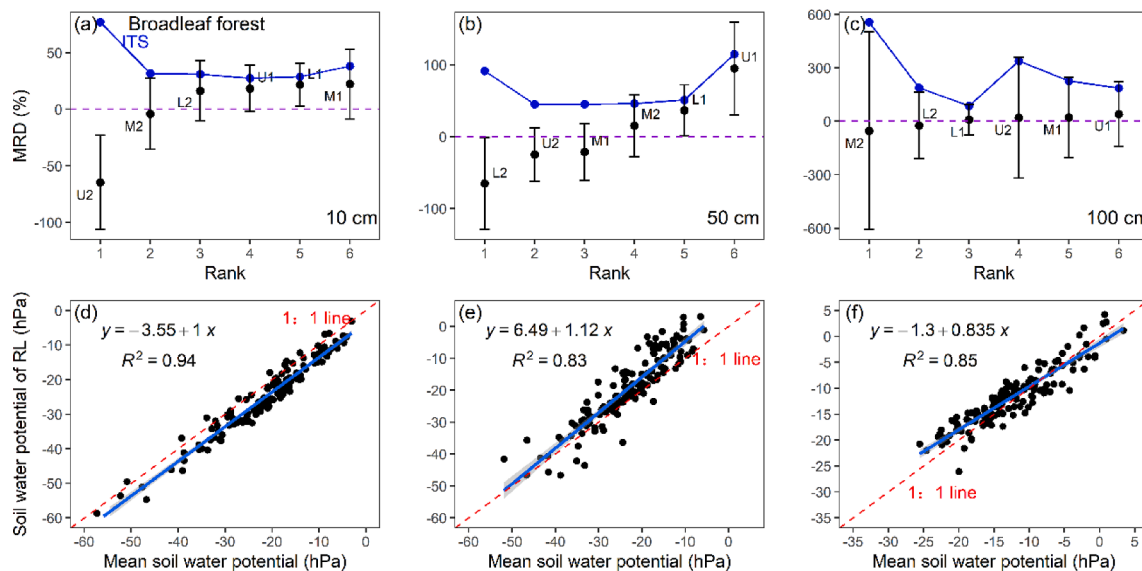


Fig. 5. Ranked MRD (with one standard deviation) for different depths (a, b, c), and comparison of the hillslope mean SWP versus SWP observed at the most representative locations (d, e, f) in the broadleaf forest.

contribution of the temporal anomalies in the broadleaf forest was higher than that in the coniferous forest at all depths, with mean values of 48.75 % (0.47 %–250.64 %) in the broadleaf forest and 24.8 % (0.18 %–73.04 %) in the coniferous forest. As a result, the covariance was highly negative, with a mean value of -41.92% (-595.28% – 49.06%) in the broadleaf forest compared with that of -2.59% (-215.54% – 46.75%) in the coniferous forest (Table 2). The negative covariance implies a negative contribution, indicating that the interactions between the factors controlling the temporal mean and those controlling the temporal anomalies are strengthened in the broadleaf forest, but such interactions are weak in the coniferous forest. As a result, the total mean contribution of the time-variant component (i.e., the sum of temporal anomalies and covariance) were 22.21 % and 6.83 % in the coniferous and broadleaf forests, indicating that the time-variant factor is visible only in the broadleaf forest.

The effects of SWP conditions on the contributions of different components were qualitatively characterized (Figs. 8–9). The

contribution of the spatial variance of temporal mean generally increased exponentially as the SWP increased, apart from at 100 cm depth in the broadleaf forest (Figs. 8–9, left column). Plots of the contribution of the temporal anomalies as a function of the SWP yielded a concave curve, and the contribution reached the minimum at a spatial mean SWP of -30 , -12 and -6 hPa (Fig. 8, middle column) and -17 , -20 and -11 hPa in the broadleaf forest (Fig. 9, middle column) from the surface to subsurface soil layers. Moreover, the distinct contribution of the covariance decreased exponentially, which was opposite to the trend of the temporal mean in the coniferous forest. It is clear that the SWP levels significantly affect the contributions of different components to the total variance. Moreover, the ranges of all components were larger in the broadleaf forest than in the coniferous forest for all soil depths (Figs. 8–9), indicative of more variability in the SWP in the broadleaf forest.

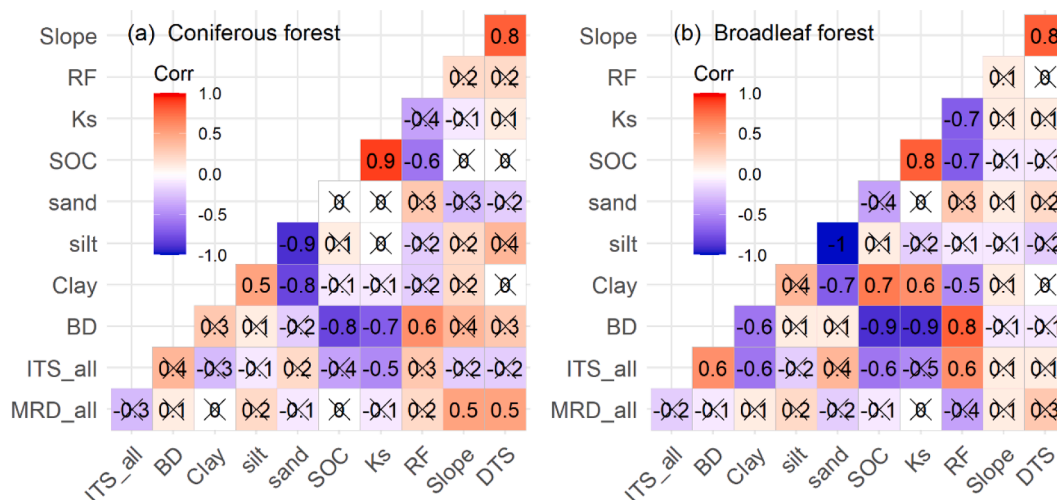


Fig. 6. Pearson correlation matrix for the temporal stability indexes (MRD and ITS) and soil terrain attributes for all depths (at significance level of 95%) in the coniferous (a) and broadleaf forests (b).

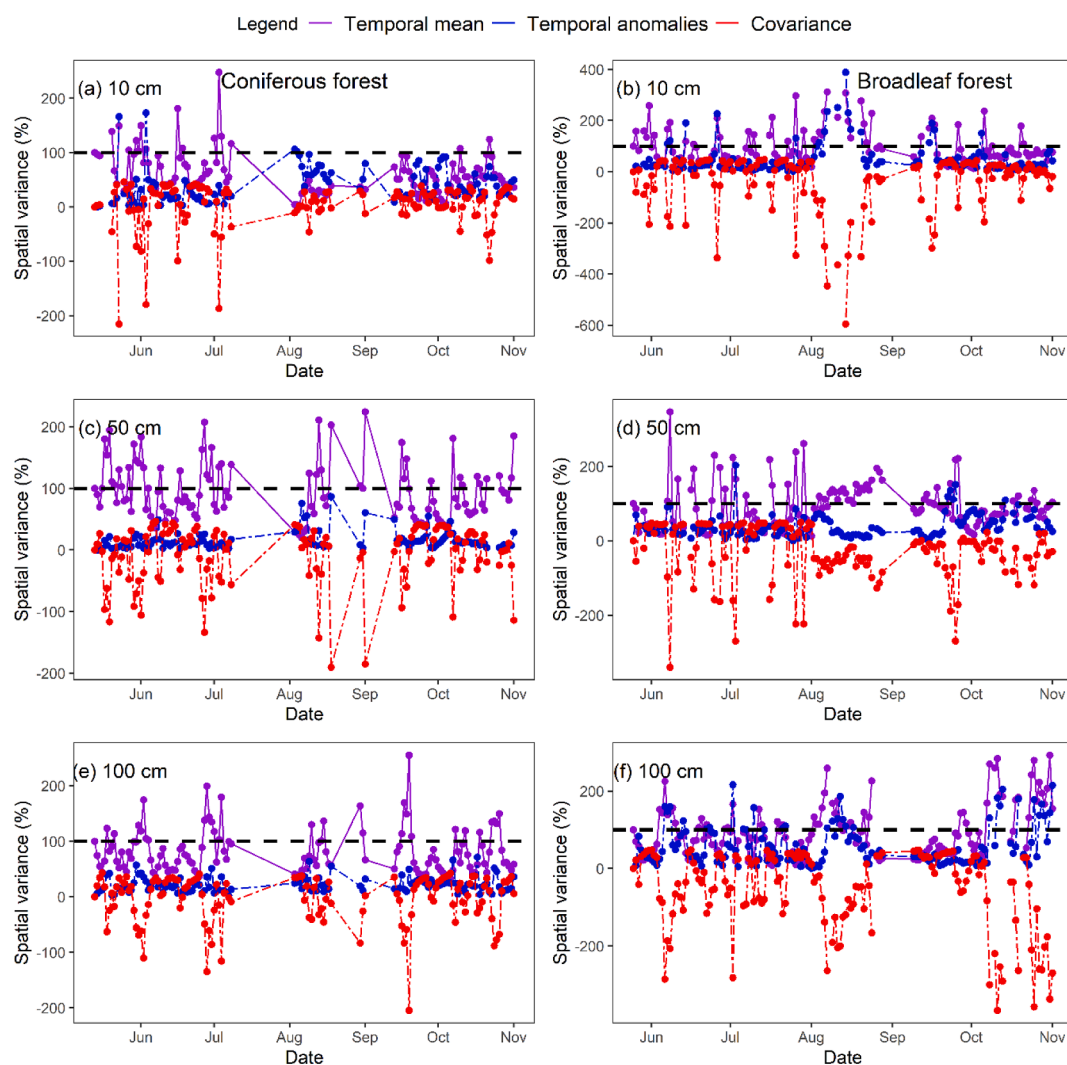
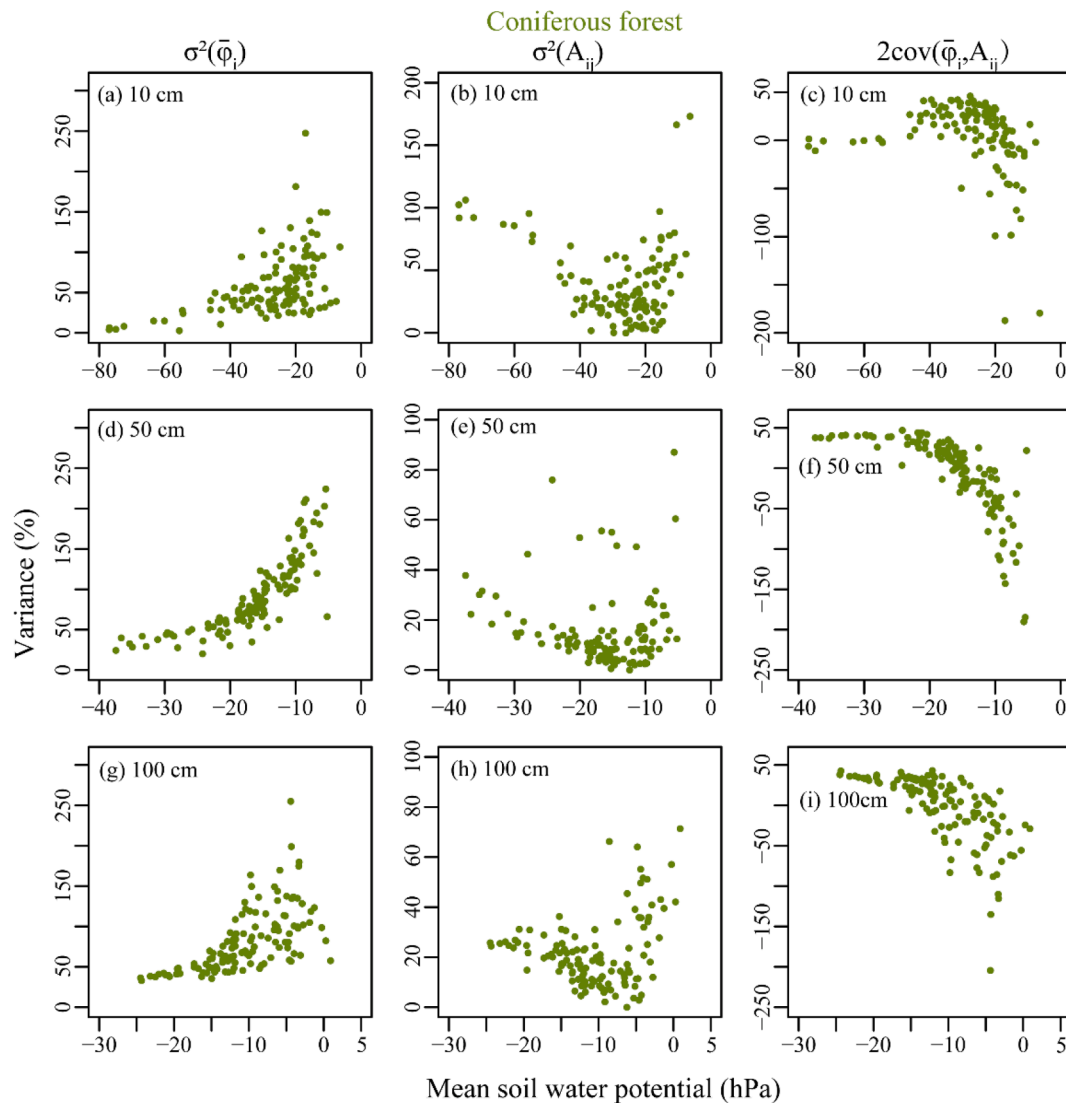


Fig. 7. Temporal evolution of spatial variance of SWP and its contribution components, expressed as the percentage of the total (black dashed line denotes the total variance 100%) in the coniferous (left column) and broadleaf forests (right column).

Table 2

Total spatial variance and its decomposition for various depths in the coniferous and broadleaf forests, expressed as the percentage of the total variance.

Decomposition of spatial variance	Coniferous forest				Broadleaf forest			
	10 cm	50 cm	100 cm	Mean	10 cm	50 cm	100 cm	Mean
Temporal mean $\sigma^2(\bar{\varphi}_i)$	59.79	92.73	80.85	77.79	92.49	91.05	95.96	93.17
Temporal anomalies $\sigma^2(A_{ij})$	37.53	15.14	21.74	24.80	48.29	39.41	58.55	48.75
Covariance $2Cov(\bar{\varphi}_i, A_{ij})$	2.69	-7.87	-2.59	-2.59	-40.78	-30.46	-54.51	-41.92
$\sigma^2(A_{ij}) + 2Cov(\bar{\varphi}_i, A_{ij})$	40.22	7.27	19.15	22.21	7.51	8.95	4.04	6.83

**Fig. 8.** The relationship between the contribution of different components and mean SWP in the coniferous forest at different depths: spatial variance of the temporal mean (a, d, and g), spatial variance of temporal anomalies (b, e, and h), and covariance (c, f, and i).

4. Discussions

4.1. Statistical and temporal stability analysis

Both hillslopes were extremely wet during the study period (Figs 1 and S1), which caused the lower spatial variability compared with the observed values in the humid Waldstein experimental site vegetated with Norway spruce in Germany, where the mean SWP was 0 to -500 hPa and the standard deviation was 0–34.0 hPa (Göttlein and Manderscheid, 1998). In the temperate Shaley Hills in central USA with deciduous forests, the SWP ranged from -9.5 to -704.9 hPa (Yu et al., 2015), where the standard deviation of the SWP was 0.4–40.4 hPa. In

the present study, a higher standard deviation was observed at 10 and 50 cm depth (i.e., the soil-root zone) in the broadleaf forest compared with the coniferous forest. One reason for this result may be the thermal conditions, because higher soil water absorption occurs in the shallow soil-root zone in the broadleaf forest (Table S1), which is equivalent to the conclusions reached in earlier studies that examined the thermal differences between north- and south-facing hillslopes (Lin, 2006; Wang et al., 2011). In addition, the high standard deviation may have been caused by the higher spatial variability of the soil saturated hydraulic conductivity in the broadleaf forest, which concurs with the findings that the higher spatial variability of saturated hydraulic conductivity leads to a higher standard deviation of the soil moisture, especially

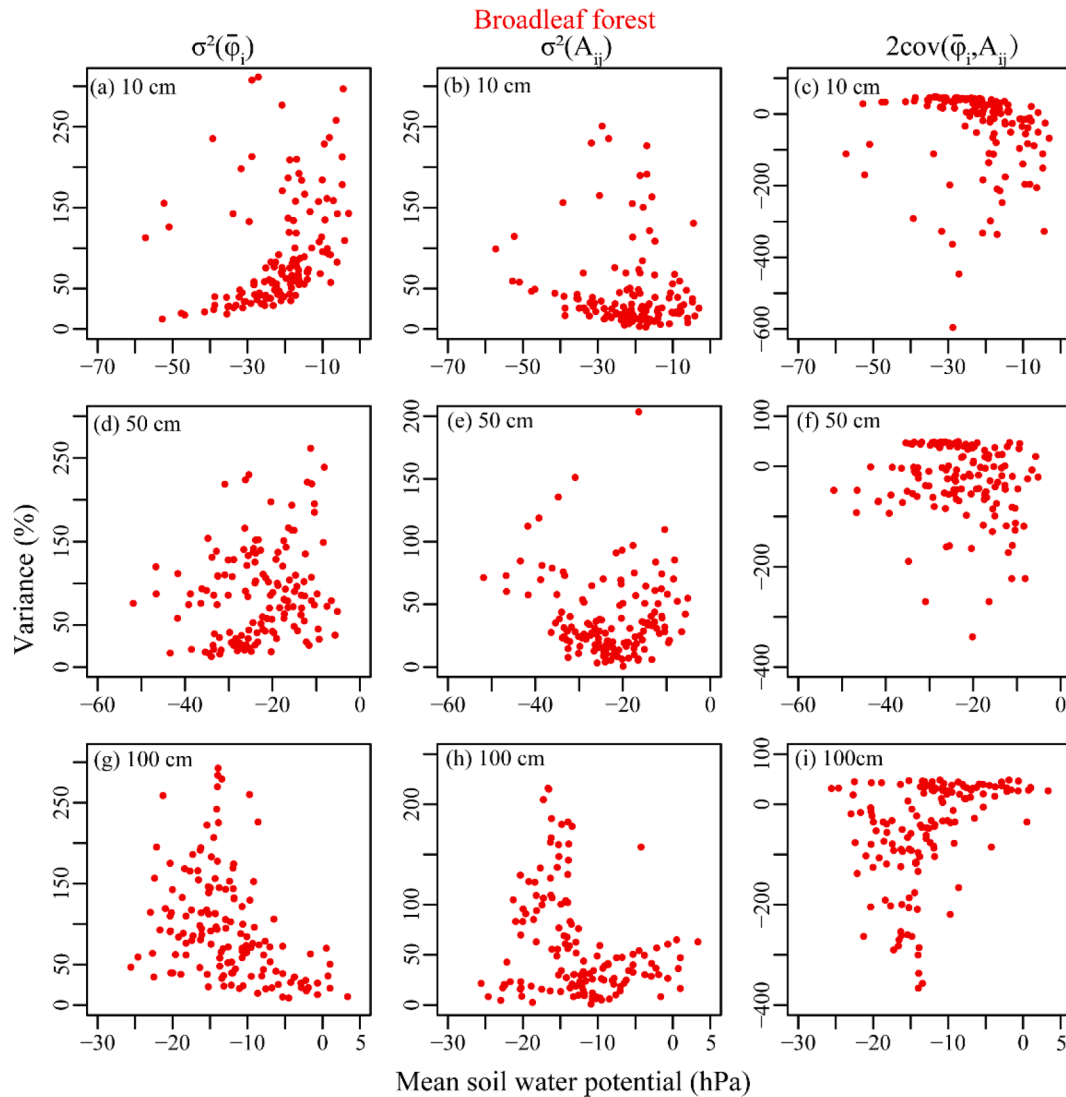


Fig. 9. The relationship between the contribution of different components and mean SWP in the broadleaf forest at different depths: spatial variance of the temporal mean (a, d, and g), spatial variance of temporal anomalies (b, e, and h), and covariance (c, f, and i).

under wet conditions (Martínez García et al., 2014; Vereecken et al., 2007). However, the soil moisture range in the form of volumetric content in the soil-root zone was smaller in the broadleaf forest (Fig. S1), which can be attributed to the inherent soil properties. The relationship between the standard deviation and $\bar{\varphi}_j$ was downward linear in both forests, with the exception at 100 cm depth in the broadleaf forest, which is consistent with the results of previous studies (Hendrickx and Wierenga, 1990; Van Pelt and Wierenga, 2001; Yu et al., 2015), suggesting that a downward linear relationship is common, with $\bar{\varphi}_j$ ranging from -500 to 0 hPa. A possible explanation for the different behavior observed at the 100 cm depth data in the broadleaf forest is that, the greater elevation difference and frequent lateral/preferential flow reduced the horizontal variance of $\bar{\varphi}_j$ (Lin, 2006; Lin and Zhou, 2008; Williams et al., 2009). This reasoning might be supported by the fact that the interquartile ranges of $\bar{\varphi}_j$ were lower in the broadleaf forest.

The temporal stability of the spatial pattern of the SWP was the highest in the soil-root zone (10 and 50 cm depths) in terms of the ITS. This result agrees with the results of previous studies conducted in tropical Piracicaba, Brazil (Cichota et al., 2006) and Shales Hills catchment, central USA (Yu et al., 2015), where the temporal stability was higher in the upper soil layers. Despite the advantages of reducing cost by choosing a single representative location, it is still challenging to

find one single representative location for the estimation of $\bar{\varphi}_j$ for different depths at the hillslope scale (Gao and Shao, 2012; Jacobs et al., 2004; Liao et al., 2017; Sur et al., 2013), meaning that temporally stable locations at different depths do not always occur at the same location. In the coniferous forest, as the soil properties on the hillslope are less variable (Table S1; i.e., the low standard error of the bulk density, clay, silt, and sand contents), the slope and DTS were identified as controlling factors on the MRD. Flatter slope and short DTS contribute to the formation of the hydrostatic pressure or water head, which was observed at the lower slope with higher standard deviation and variability in the coniferous forest (Fig. 2a), whereas the steeper slope and relatively drier SWP caused by long DTS may drain infiltration water rapidly to the deeper soil layers or lateral direction, thus affecting SWP variability (Gao et al., 2013; Jarecke et al., 2021; Kaiser and McGlynn, 2018; Martinez et al., 2013). Therefore, representative sampling/monitoring locations should be a moderate distance from streams and on average slope in the coniferous forest, which is consistent with the finding of previous studies (Brocca et al., 2009; He et al., 2019; Sur et al., 2013). No factors were significantly related to the MRD in the broadleaf forest, which may be because of the extreme soil heterogeneity caused by the dense root networks and abundant rock fragments that result in high spatial variability of the SWP. Meanwhile, intrinsically related soil

properties (i.e., soil bulk density, SOM, clay and rock fragment content) were identified as controlling factors on the ITS. The rock fragments increase the soil bulk density and indicate weak weathering conditions, lead to low clay content, which has been proven to be a key factor influencing temporal stability characteristics (Jacobs et al., 2004; Sur et al., 2013). The rock fragments provide preferential flow pathways for infiltrating water by changing the soil hydraulic conductivity, which is thought to significantly affect the spatial variability of the SWP (Martínez García et al., 2014; Vereecken et al., 2007).

4.2. Contribution to the total variance

The spatial variance of the time-invariant component varies with the forest types and soil depths (Table 2). At the surface 10 cm soil depth, the contribution from the time-invariant component was 2.66 %–247.53 % with a mean value of 59.79 % in the coniferous forest and 12.38 %–311.32 % with a mean value of 92.49 % in the broadleaf forest. These values are much larger than the results from the study of Brocca et al. (2014), who found that the time-invariant component contributed to the total variance by 38 %–117 % with a mean value of 33 % in six monitoring networks at a depth of 5–15 cm at sites covered with grass and cropland. However, our results are similar to those from a study conducted in a loess catchment vegetated with perennial grasses, where the time-invariant component contribution varied between 21.4 % and 282.2 % with a mean value of 66.8 % in the surface soil layer (Gao et al., 2015). Gao et al. (2015) attributed the greater contribution of the time-invariant component to the increased spatial variability of soil properties caused by the complex topography. For instance, more fine soil particles are deposited at lower elevations and slope gradients. In our study, the soil properties along the hillslopes are similar, and the differences between the forest types were 32.70 %, 10.94 %, and 43.47 % in the surface soil layer for the temporal mean, temporal anomalies, and covariance, respectively. A possible explanation for the differences is complex soil structures and evapotranspiration. The variability is mainly controlled by hydraulic conductivity in wet conditions (Korres et al., 2015; Vereecken et al., 2007), and we found larger and more variable K_s in the broadleaf forest than in the coniferous forest due to the denser root network (Table S1). This significant difference in K_s is comparative with the previous study that rice and wheat crop cover showed a similar contribution of the time-invariant component due to a minor difference of soil hydraulic properties (Yetbarek and Ojha, 2020). The higher evapotranspiration caused by lower elevation in the broadleaf forest leads to an increase in variability, leading to an increase in higher contribution from the time-invariant component (Famiglietti et al., 1998; Korres et al., 2015). In the subsurface soil layers, the temporal changes in the time-invariant (i.e., temporal mean) and time-variant (i.e., temporal anomalies plus covariance) components were similar to those observed in the surface soil layer, which has also been recognized in previous studies (Mittelbach and Seneviratne, 2012; Yetbarek and Ojha, 2020). The time-invariant component is still the predominant contributor with average values exceeding 80 % in all subsurface soil layers (i.e., 50 and 100 cm depth) in both forests, which is higher than the average value of 60 % in subsurface soil layers (i.e., 40, 60, and 80 cm depth) in the loess catchment (Gao et al., 2015). This may be attributed to the coarse soil texture, which is mainly characterized by abundant rock fragments, thus leading to higher variability of the SWP due to the preferential flow (Wang et al., 2022).

In our study area, the soil wetness clearly affects the contributions of the components, and is more pronounced in the coniferous forest (Figs. 8–9). The contribution of the temporal mean generally exponentially increased with increasing mean SWP, and that of the temporal anomalies became near zero at intermediate SWP conditions, which is consistent with previous studies (Gao et al., 2015). When there is rainfall input, the moist soil quickly becomes near-saturated, where the static properties (i.e., soil hydraulic conductivity) take the controlling role of water flux and determine the variability of the SWP. The contribution of

the temporal anomalies decreased with increasing soil wetness until a threshold was reached, at which point connected flow paths and lateral flow dominated the soil hydrological processes resulting in an increased contribution of temporal anomalies (Grayson et al., 1997; Jarecke et al., 2021).

Moreover, the spatial scale is another important aspect to consider for the analysis. In general, soil properties and vegetation are expected to be less variable at a smaller spatial scale, and the dynamic properties are expected to be dominant in controlling the spatial variance of the SWP. A decreasing trend in the contribution of the time-invariant component to the total spatial variance was observed from 94 % over an area of 31,500 km² (Mittelbach and Seneviratne, 2012) to 66.8 % over an area of 0.6 km² (Gao et al., 2015). Therefore, the contribution of the time-invariant component at small scales is expected to be lower (Brocca et al., 2014; Gao et al., 2015). The unexpected dominant contribution of the time-invariant component in this study may be due to the extremely wet soil conditions, when nonlocal factors have a dominant control on soil moisture patterns by governing lateral water distribution through subsurface paths on the basis of the recognized two preferred states (i.e., lateral water movement dominates under the wet-preferred state, whereas vertical movement, such as evapotranspiration and vertical drainage, dominates under the dry-preferred state) of soil moisture (Famiglietti et al., 1998; Grayson et al., 1997). This result is consistent with Yetbarek and Ojha (2020), who also found a more significant contribution from the time-invariant component in areas with 0.0006 km². Therefore, the contradictory results in the present study indicate that the effects of complex subsurface flow induced by static properties (e.g., soil depth, soil structures, and subsurface layering) on the soil moisture/water potential dynamics require further investigations.

5. Conclusions

In this study, we characterized variability and temporal stability of SWP at different soil depths under coniferous and broadleaf forested hillslopes located on the eastern edge of the Tibetan Plateau. The SWP showed a higher variability and lower temporal stability in the soil–root zone (i.e., 10 and 50 cm depths) of the broadleaf forest owing to the higher evapotranspiration and more spatially variable soil saturated hydraulic conductivity compared with the coniferous forest. The temporal stability of SWP at the deeper soil layer (100 cm depth) in both forests was higher than the soil–root zone soil depths. The correlation analysis between temporal stability indexes and related attributes revealed that the topography (i.e., slope and DTS) in the coniferous forest and soil properties (i.e., bulk density, SOC, clay and rock fragments contents) in the broadleaf forest were identified as the controlling factors on the temporal stability of SWP in the hillslopes, indicating that the variability of SWP was attributed to a complex interaction between topography, soil properties, vegetation, and elevation. These results represent a useful support to optimize soil moisture monitoring/sampling schemes over areas at the hillslope scale.

Overall, the time-invariant component (temporal mean) was the major contribution (59.79–92.73 % in the coniferous forest and 91.05–95.96 % in the broadleaf forest) to the total spatial variance of the SWP despite the minor difference in soil properties and continuous rainfall input at the studied scale and region. The time-variant component (temporal anomalies plus covariance) was more visible in the coniferous forest (7.27–40.22 %) than the broadleaf forest (4.04–8.95 %). The unexpected high time-invariant contribution is mainly attributed to the soil spatial structures relating to the preferential flow and the subsurface lateral flow caused by the interaction of the vegetation, topography, and climate condition in this study. Soil wetness conditions significantly affect the contribution of individual components. In general, there is an exponential increase between the contribution of the temporal mean and hillslope mean SWP in both forests, and a concave shape between the contribution of the temporal anomalies component

and hillslope mean SWP was also found, although this relationship is not so pronounced in the subsurface layer in the broadleaf forest.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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